3.2.5 The Loc/Wec/Plant<N>/Para node

The *Para* node can be used to view the control parameters of a wind energy converter or an ENERCON SCADA device. The item designation corresponds to the parameter designation in the control system or the ENERCON SCADA device.

The *Loc/Wec/Plant<N>/Para/User* node also includes items whose designations correspond to those of the parameters in the control system or the ENERCON SCADA device. The value of these items indicates who (user) made the last parameter change.

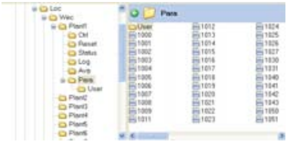

Fig. 14: The *Loc/Wec/Plant<N>/Para* node

CS48a

The following parameters are available for wind energy converters with a **CS48a** control system:

Item	Description	Unit
1000	Startdelay:	s
1001	nominal mains volt.:	V
1002	mains undervoltage level U<:	%
1003	tripping time during undervoltage U<:	s
1004	mains undervoltage level U<<:	%
1005	tripping time during undervoltage U<<	s
1006	mains overvoltage level U>:	%
1007	tripping time during overvoltage U>:	s
1008	mains overvoltage level U>>:	%
1009	tripping time during overvoltage U>>:	s
1011	nominal mains freq.:	Hz
1012	mains underfrequency level:	Hz
1013	tripping time during underfrequency:	s
1014	mains overfrequency level:	Hz
1015	tripping time during overfrequency:	s
1016	frequency control:	
1017	frequency control referred to:	
1018	way of frequency control:	

Item	Description	Unit
1019	return frequency:	Hz
1020	limitation frequency 1:	Hz
1021	limitation frequency 2:	Hz
1022	limitation frequency 3:	Hz
1023	limitation frequency 4:	Hz
1024	limitation frequency 5:	Hz
1025	reserved power (f2-f3)	%
1026	limit. power (f4):	%
1027	limit. power (f5):	%
1030	grid type:	
1031	FACTS-operation:	
1040	mains undervoltage level for restart	%
1041	mains overvoltage level for restart	%
1042	mains underfrequency level for restart	Hz
1043	mains overfrequency level for restart	Hz
1050	inertia control:	
1051	inertia control:mode:	
1052	inertia trig. freq.:	Hz
1053	inertia min. freq.:	Hz
1054	inertia factor:	%
1055	max. inertia time:	s
1056	inertia recov. time:	s
1100	UVRT-mode during unsym. grid faults:	
1101	UVRT-mode during symmetr. grid faults:	
1102	UVRT trigger voltage:	%
1103	UVRT return voltage:	%
1104	UVRT trigger voltage ZPM:	%
1105	OVRT trigger voltage:	%
1106	OVRT return voltage:	%
1107	max. UVRT time during unsymmetr. faults:	s
1108	max. UVRT time during symmetr. faults:	s
1109	max. OVRT time:	s
1114	react. power gradient after ZPM:	MVAr/s
1115	active power gradient after ZPM:	MW/s

Item	Description	Unit
1130	mains contactor during ZPM:	
1200	Statcom operation:	
1201	Farm Control Interf.:	
1202	apparent power limitation by:	
1203	v-curve gradient Q(rat) at P/P(rat):	%
1204	rated reactive power:	kvar
1205	reserved reactive power:	kvar
1209	dyn. react.pow.(Q/P): (farm contr. fault)	
1210	static react. pow.: (farm contr. faulty)	kvar
1308	SCADA farm control:	
1309	dP/dt-control:	
1311	storm control:	
1314	blade heating: automatic:	
1315	permit blade heating during operation:	
1316	max. power consump.of blade heating:	kW
1317	min.heating duration of blade heating:	min
1400	maximum power (farm contr. faulty):	kW
2000	site height above sealevel:	m
2003	nominal power:	
2010	cold climate package:	
2100	maximum power (power optimized):	kW
2101	maximum power (noise optimized):	kW
2102	power gradient (mains breakdown):	kW/s
2103	power gradient (during start):	kW/s
2104	power gradient (during operation):	kW/s
2501	nominal speed (noise optimized):	rpm
3101	Labko ice detector	
3102	Labko ice detector: automatic restart aft	
4011	sectoral limitation: start angle sector 1	°
4012	sectoral limitation: start angle sector 2	°
4013	sectoral limitation: start angle sector 3	°
4014	sectoral limitation: start angle sector 4	°
4015	sectoral limitation: start angle sector 5	°
4016	sectoral limitation: start angle sector 6	°

Item	Description	Unit
4017	sectoral limitation: start angle sector 7	°
4018	sectoral limitation: start angle sector 8	°
4021	sectoral limitation: end angle sector 1	°
4022	sectoral limitation: end angle sector 2	°
4023	sectoral limitation: end angle sector 3	°
4024	sectoral limitation: end angle sector 4	°
4025	sectoral limitation: end angle sector 5	°
4026	sectoral limitation: end angle sector 6	°
4027	sectoral limitation: end angle sector 7	°
4028	sectoral limitation: end angle sector 8	°
4031	sectoral limitation: min. wind speed for	m/s
4032	sectoral limitation: min. wind speed for	m/s
4033	sectoral limitation: min. wind speed for	m/s
4034	sectoral limitation: min. wind speed for	m/s
4035	sectoral limitation: min. wind speed for	m/s
4036	sectoral limitation: min. wind speed for	m/s
4037	sectoral limitation: min. wind speed for	m/s
4038	sectoral limitation: min. wind speed for	m/s
4041	sectoral limitation: max. wind speed for	m/s
4042	sectoral limitation: max. wind speed for	m/s
4043	sectoral limitation: max. wind speed for	m/s
4044	sectoral limitation: max. wind speed for	m/s
4045	sectoral limitation: max. wind speed for	m/s
4046	sectoral limitation: max. wind speed for	m/s
4047	sectoral limitation: max. wind speed for	m/s
4048	sectoral limitation: max. wind speed for	m/s
4051	sectoral limitation: max. power in sector	kW
4052	sectoral limitation: max. power in sector	kW
4053	sectoral limitation: max. power in sector	kW
4054	sectoral limitation: max. power in sector	kW
4055	sectoral limitation: max. power in sector	kW
4056	sectoral limitation: max. power in sector	kW
4057	sectoral limitation: max. power in sector	kW
4058	sectoral limitation: max. power in sector	kW

Item	Description	Unit
4061	sectoral limitation: min. blade angle in	°
4062	sectoral limitation: min. blade angle in	°
4063	sectoral limitation: min. blade angle in	°
4064	sectoral limitation: min. blade angle in	°
4065	sectoral limitation: min. blade angle in	°
4066	sectoral limitation: min. blade angle in	°
4067	sectoral limitation: min. blade angle in	°
4068	sectoral limitation: min. blade angle in	°
4091	sectoral limit.: power grad. sector 1-8	kW/s
4106	limit for farm ice detection:	%
4107	automatic restart after icing:	
4108	automatic restart during icing:	
4109	position nacelle during icing up to 7m/s	
4110	nacelle position at icing:	°
6000	noise optimization	
6001	minimum time	min
6002	noise optimization time	
6003	time from	hh:mm
6004	time to	hh:mm
6005	noise optimization wind speed	
6006	Delay	min
6007	min. wind speed	m/s
6008	max. wind speed	m/s
6009	noise optimization wind direction	
6010	delay	min
6011	angle 1. sector	°
6012	range 1. sector	°
6013	angle 2. sector	°
6014	range 2. sector	°
6015	angle 3. sector	°
6016	range 3. sector	°
6017	combination	
6100	shadow shut off	
6101	Shut off intensity	%

Item	Description	Unit
6102	filter on	s
6103	filter off	s